

2 A group of ecologists investigated the biodiversity of two peat bog ecosystems, **A** and **B**.

Peat bogs are wetland ecosystems that often contain rare species.

Fig. 2.1 shows an example of a peat bog.



Fig. 2.1

(a) The ecologists sampled the plant species in peat bog **A** and peat bog **B**.

The ecologists decided to use two suitable indices of biodiversity:

- Simpson's index of diversity
- Shannon diversity index.

The ecologists had read that the two indices can lead to different conclusions.

The Shannon diversity index gives values in the range 1.5–3.5, where 1.5 is low biodiversity and 3.5 is high biodiversity.

The results are shown in Table 2.1.

(b) The ecologists sampled the invertebrate species in peat bogs **A** and **B** and calculated invertebrate biodiversity using Simpson's index of diversity.

(i) The ecologists sampled invertebrate species in peat bog **A**:

- between 13:00 and 16:00 on three different days within a 30-day period
- at 10 random sites along the edge of a path in the peat bog
- by sampling the invertebrates within a 1 m² quadrat placed at each site
- by using an identification key to identify the species
- by counting the number of individuals of each species.

Describe how this sampling method could be improved to make sure the results are more representative of all invertebrates in the whole of peat bog **A**.

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- (ii) Table 2.2 shows the results of the invertebrate sampling in peat bog **A**.

The formula for Simpson's index of diversity (D) is:

$$D = 1 - \left(\sum \left(\frac{n}{N} \right)^2 \right)$$

n = number of individuals of each species present in the sample

N = total number of all individuals of all species present in the sample

Complete Table 2.2 **and** use the formula provided to calculate Simpson's index of diversity.

Table 2.2

species	n	n/N	$(n/N)^2$
<i>Chartoscirta cocksii</i>	11	0.071	0.005
<i>Eristalis cryptarum</i>	72	0.462	0.213
<i>Glyphesis cottonae</i>	8	0.051	0.003
<i>Glyphesis servulus</i>	6	0.038	0.001
<i>Loxocera nigrifrons</i>	29	0.186	0.035
<i>Stenus argus</i>	10	0.064	0.004
<i>Trechus rivularis</i>	7	0.045	0.002
<i>Tipula limbata</i>	13		
$\Sigma =$			

Simpson's index of diversity (D) = [3]

- (iii) The ecologists calculated a D value of 0.710 for peat bog **B**.

Use your value of D from (b)(ii) to compare the **biodiversity** of peat bog **A** and peat bog **B**.

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 [1]

[Total: 9]